NATIONAL UNIVERSITY OF SINGAPORE
INTELLIGENT REASONING SYSTEMS

PROJECT REPORT

HYBRID MACHINE REASONING FOR CYBER-PHYSICAL SECURITY

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1 Introduction

Organizations rely on both cybersecurity and physical security systems to protect their information assets, infrastructure, and facilities. Cybersecurity systems monitor digital activities such as authentication attempts, file access, and removable storage device usage, while physical security systems record activities such as employee access to buildings, offices, and restricted locations.

Although these systems contribute to organizational security, the events they generate are often analyzed in isolation, even though security threats may involve both physical and digital domains. For example, an employee accessing a restricted facility outside normal working hours may not independently indicate malicious behavior. However, when this activity occurs close to suspicious cybersecurity events associated with the same employee, the combined evidence may indicate an elevated security risk. Analyzing these events separately may prevent organizations from identifying such cross-domain relationships.

Furthermore, identifying suspicious behavior requires more than detecting individual anomalies. Security assessments should consider how multiple events are related and whether their combined characteristics indicate potential threats. Therefore, an integrated approach is needed to represent relationships between cyber and physical security events, apply reasoning techniques to interpret these relationships, and produce explainable security risk assessments.

The aim of this project is to develop and evaluate a hybrid machine reasoning system that correlates cyber and physical security events to identify potential cyber-physical security risks that may not be apparent when each domain is analyzed independently.

The project focused on the following objectives:

- **Improve security risk detection:**
  Identify potential cyber physical security risks by analyzing cyber and physical access events together, rather than treating them as separate incidents.
- **Provide contextual security insights:**
  Reveal meaningful relationships between users, devices, locations, and timestamps to detect suspicious patterns that may otherwise be overlooked.
- **Support proactive risk management:**
  Classify scenarios into Low, Medium, or High Risk to help security teams prioritize cases requiring attention or further investigation.
- **Provide transparent and explainable decisions:**
  Generate evidence-based explanations showing the relevant events, relationships, and rules that contributed to each risk classification.
- **Enhance analyst decision making:**
  Support security analysts with a combined rule-based and machine learning assessment, while keeping human review central for ambiguous or conflicting cases.

The developed framework integrates knowledge representation, symbolic reasoning, and machine learning to analyze cyber-physical security events within a unified environment. A Knowledge Graph (KG) was constructed to represent organizational users, devices, physical locations, and security events, together with the relationships connecting them. Symbolic reasoning rules were implemented to infer contextual information and identify potentially suspicious combinations of activities.

In parallel, machine learning classification techniques, including a Decision Tree, were implemented to predict security risk levels using features derived from the integrated data. The outputs of the symbolic reasoning and machine learning components were combined to support the final security risk assessment.

The framework also provides human-readable explanations describing the evidence, relationships, and reasoning outcomes contributing to each assessment. Its effectiveness was evaluated against predefined ground-truth scenarios using classification performance metrics and explanation-quality criteria. Additionally, cyber-only, physical-only, and integrated classification approaches were compared to examine the contribution of combining security information across both domains.

2 Literature Review

2.1 Scope and analytical perspective

This review examines how relationships between organizational activities can be represented, correlated and interpreted to support security risk assessment. The focus is not the broad field of industrial cyber-physical systems, but the convergence of physical access-control events (such as entry to a restricted room) and cybersecurity events (such as failed logins, sensitive-file access and removable-media use). The proposed project represents users, assigned devices, locations and time-stamped events in a knowledge graph (KG), applies explicit contextual rules, and combines symbolic and machine-learning outputs to classify scenarios as low, medium or high risk. Accordingly, the literature is synthesized around four connected questions: what graphs contribute to security analysis; what has been achieved by converging physical and logical monitoring; how heterogeneous events can be correlated; and what hybrid and explainable reasoning adds beyond detection alone.

2.2 Knowledge graphs in cybersecurity

Knowledge Graphs (KGs) represent entities and their relationships explicitly, enabling analysts to explore connections that may not be readily visible in isolated records. Alharbi et al. (2025) constructed a cybersecurity Knowledge Graph by integrating heterogeneous structured and unstructured data sources, enriching the graph through logical rules and evaluating its ability to retrieve previously implicit relationships between security entities. Their work demonstrates how logical rules can infer additional relationships, extending the graph beyond the information explicitly represented in the original data. However, their approach primarily focuses on cybersecurity knowledge, including vulnerabilities and weaknesses, rather than correlating physical access and cybersecurity activities associated with organizational users.

Li et al. (2025) investigated the construction of cybersecurity Knowledge Graphs from cyber threat intelligence reports. Their CyberKG framework integrates domain-specific language models with sequence modeling and attention mechanisms to extract cybersecurity entities and relationships. The study demonstrates how unstructured threat intelligence can be transformed into structured entities and relationships within a Knowledge Graph. However, its primary contribution concerns information extraction and Knowledge Graph construction. In contrast, the present study uses structured organizational, physical access, and cybersecurity event records, emphasizing relationship modeling and contextual security analysis rather than natural language processing for information extraction.

Loevenich et al. (2025) explored the integration of cybersecurity Knowledge Graphs, large language models, and automated planning techniques to generate attack chains. Their proposed approach illustrates the potential of graph-based knowledge representation to support automated attack-chain reasoning and planning. However, the study primarily addresses cyber threat intelligence and attack-

chain generation rather than the correlation and risk classification of employee physical access and cybersecurity activities.

Taken together, these studies demonstrate the value of Knowledge Graphs in representing interconnected security information and supporting contextual analysis through graph-based queries. In the present study, the Knowledge Graph represents relationships between users, devices, physical access events, and cybersecurity activities. Cypher queries are used to retrieve and analyze these relationships, identify security-relevant conditions such as after-hours activity and restricted-location access, and correlate physical and cyber events based on shared entities and temporal proximity. These graph-based analyses produce contextual security indicators and supporting evidence, which are subsequently incorporated into symbolic reasoning and machine-learning components to support the overall security risk assessment.

2.3 Convergence of physical and cybersecurity monitoring

Fausto et al. (2021) proposed a logical architecture that integrates physical security and cybersecurity information to improve anomaly detection in critical infrastructure environments. Their study evaluated machine-learning techniques for identifying suspicious activities through the correlation of physical and digital security events. The findings demonstrate the value of combining information from both domains, particularly when suspicious patterns may not be identifiable through isolated monitoring. This research is closely related to the present study because it considers physical access activities alongside cybersecurity events. However, its approach primarily focuses on machine-learning-based anomaly detection rather than Knowledge Graph-based contextual analysis and hybrid reasoning for security risk assessment.

Aloufi et al. (2025) examined the integration of cybersecurity and physical security within healthcare institutions, emphasizing access control, surveillance, unified monitoring, and organizational coordination. Their review highlights the importance of integrated security approaches in improving situational awareness and addressing threats that may affect both physical and digital environments. However, the study primarily provides a conceptual and organizational perspective rather than implementing and experimentally evaluating an integrated security-event correlation and reasoning framework.

Together, these studies demonstrate the importance of integrating physical and cybersecurity information to provide a more comprehensive understanding of security threats. While the convergence of these domains has already been investigated, the present research focuses on extending this integration through Knowledge Graph-based contextual analysis, temporal event correlation, and hybrid intelligent reasoning. By representing relationships between users, devices, locations, and security events, the proposed framework aims to support explainable and evidence-based risk assessments that consider suspicious activities across both physical and cybersecurity domains.

2.4 Security-event correlation and contextual interpretation

Coppolino et al. (2016) proposed the Generic Event Translator (GET), a framework designed to integrate and pre-correlate heterogeneous security information across physical and logical security layers. Integrated with Security Information and Event Management (SIEM) systems, the framework incorporates contextual information to support the interpretation of security events and was evaluated in dam-control and mobile payment scenarios. This study demonstrates the importance of integrating security information from different sources and establishing consistent event representations to support meaningful correlation. It also highlights the need to identify relationships between events and their associated entities when analyzing security activities across multiple systems.

Shaukat et al. (2026) reviewed event-log correlation techniques for detecting multi-step cybersecurity attacks, examining approaches including rule-based, statistical, sequence-based, graph-based, causal, and hybrid methods. Their review highlights the importance of selecting correlation techniques according to the characteristics of security events and detection objectives, while considering challenges such as temporal relationships, heterogeneous data sources, and false-positive alerts. Although the study primarily focuses on cybersecurity event logs, its findings provide a useful foundation for understanding the challenges associated with correlating security information from multiple sources.

Building on these approaches, the present study represents correlated physical-access and cybersecurity events within a Knowledge Graph using explicit relationships between users, assigned devices, and security events. Cypher queries traverse these relationships to retrieve cross-domain contextual information and calculate temporal associations between the paired events. Additional contextual attributes, including normal working hours, restricted-location access, and access outcomes, are incorporated to support the subsequent symbolic reasoning components.

2.5 Intelligent and explainable security reasoning

Mehmmod et al. (2025) proposed an enhanced rule-based intrusion detection model that combines fuzzy logic with machine-learning classifiers to identify suspicious network activity in IoT environments. Their experimental results demonstrate the potential benefits of integrating explicit rule-based processing with data-driven classification. However, the approach primarily operates on network-traffic features rather than relationships between organizational users, physical access events, and cybersecurity activities. Therefore, while the study supports the use of hybrid detection techniques, its reported performance cannot be directly generalized to cross-domain security risk assessment.

Nazir et al. (2026) investigated explainable intrusion detection and alert prioritization by combining an XGBoost-based detection model with SHAP and LIME explanation techniques. Their framework was evaluated using the CICIoT2023 and UNSW-NB15 datasets. Interestingly, their findings indicate that incorporating additional temporal and graph-based contextual components did not consistently improve detection performance compared with the XGBoost baseline. The researchers attributed these limitations partly to the absence of relevant host and temporal identifiers in the evaluated data. These findings highlight that combining multiple analytical techniques does not necessarily improve detection accuracy and that the availability of meaningful contextual information is important when evaluating relational approaches.

Furthermore, Nazir et al. (2026) demonstrate the importance of distinguishing machine-learning explanations from evidence-based security reasoning. Techniques such as SHAP and LIME explain the

contribution of individual features to model predictions, whereas Knowledge Graph-based analysis can provide traceable relationships between security entities and events. In the present study, explainability is approached through the identification of relevant security events, contextual relationships, and reasoning conditions that contribute to the overall risk assessment. This enables security analysts to examine the evidence supporting a classification without relying solely on the predicted risk level.

Aalizadeh and Raahemi (2026) conducted a systematic literature review of Knowledge Graphs and semantic technologies in cybercrime investigation and digital forensics. Their study examines approaches involving ontologies, logical reasoning, and hybrid neuro-symbolic architectures for representing heterogeneous information and supporting interpretable analysis. Although their research primarily focuses on digital investigations rather than integrated physical and cybersecurity risk assessment, its emphasis on knowledge representation, evidence traceability, and explainable reasoning provides a relevant foundation for the proposed framework.

2.6 Synthesis, research gap and implications for the proposed framework

The reviewed literature demonstrates significant advancements in the application of Knowledge Graphs, security-event correlation, and intelligent reasoning techniques for cybersecurity. Existing studies have explored the integration of physical and cybersecurity monitoring, the correlation of heterogeneous security events, and the use of machine learning and symbolic reasoning to improve threat detection. However, while these approaches provide valuable contributions individually, the reviewed literature offers limited evidence of an integrated framework that combines Knowledge Graph-based contextual analysis, cross-domain security-event correlation, and hybrid intelligent reasoning to support explainable security risk assessment.

To address this gap, the proposed research investigates a hybrid reasoning framework that integrates physical access and cybersecurity events within a unified Knowledge Graph. The graph represents relationships between users, devices, locations, and security events, while Cypher queries analyze these relationships to identify relevant security conditions, including after-hours activities, restricted-location access, and temporally correlated cyber and physical events. The resulting contextual information supports symbolic reasoning and machine-learning techniques, which contribute to classifying security scenarios into Low, Medium, and High Risk categories. By combining these approaches, the framework aims to assess security risks using evidence derived from multiple security domains rather than relying on isolated events.

Furthermore, the reviewed studies emphasize the importance of evaluating intelligent security systems beyond classification accuracy alone. Effective security risk assessment requires not only identifying suspicious activities but also providing transparent and verifiable explanations of the factors contributing to each decision. Accordingly, the proposed framework emphasizes evidence-based explainability by linking risk classifications to the underlying security events, Knowledge Graph relationships, and reasoning conditions. This approach aims to support security analysts in understanding and validating security assessments while improving the interpretation of potentially suspicious activities across physical and cybersecurity domains.

3 Dataset

3.1 Dataset Overview

The dataset used in this project integrates information regarding organizational users, physical access-control activities, and cybersecurity events. Due to privacy and security constraints surrounding real corporate data, synthetic data was built upon an open-source foundation to generate this dataset. The resulting data includes records for approximately 50 users, 800 physical access events, and 800 cybersecurity events.

The dataset comprises the following components:

Users dataset:

This dataset contains relatively static information about each employee and provides contextual information for reasoning about their activities.

Table 1 - Users Dataset Fields

Field	Description
user_id	Unique identifier assigned to each user.
role	Organizational role of the user, such as Software Engineer or IT Administrator.
department	Department to which the user belongs.
device_id	Unique identifier of the device assigned to the user.
normal_start	Expected starting time of the user's normal working hours.
normal_end	Expected ending time of the user's normal working hours.

Physical event dataset:

This dataset represents physical access-control events occurring at organizational locations.

Table 2 - Physical Events Dataset

Field	Description
event_id	Unique identifier assigned to each physical access event.
user_id	Identifier of the user associated with the physical access event.
timestamp	Date and time when the physical access event occurred.
location	Physical location where the access attempt occurred, such as an office or server room.
location_type	Classification of the location as Normal or Restricted.
access_result	Outcome of the physical access attempt, either Granted or Denied.

Cyber event dataset:

This dataset represents cyber activity associated with users and their assigned devices.

Table 3 - Cyber Events Dataset

Field	Description
event_id	Unique identifier assigned to each cybersecurity event.
user_id	Identifier of the user associated with the cyber event.
device_id	Identifier of the device involved in the cyber event.
timestamp	Date and time when the cyber event occurred.
failed_logins	Number of unsuccessful authentication attempts associated with the event.
files_accessed	Total number of files accessed during the recorded activity.
sensitive_files	Number of sensitive files accessed during the recorded activity.
usb_connected	Indicates whether a USB device was connected (1 = Yes, 0 = No).
severity	Severity level assigned to the cyber event, classified as Low, Medium, or High.

3.2 Dataset pre-processing

To prepare the datasets for Machine Learning classification, Knowledge Graph construction, and symbolic reasoning, several preprocessing steps are performed. These include checking for missing values, validating user and device identifiers, removing duplicate records, and standardizing timestamp formats. The three datasets are integrated using shared user identifiers and associated device information. Cyber and physical events are correlated based on their temporal proximity within a predefined time window. Additional features derived from the original attributes. These new features include:

- **After-hours activity:** Determined by comparing event timestamps with the user's normal working schedule.
- **Restricted-location indicator:** Derived from the classification of the accessed physical location.
- **Access-denied indicator**: Derived from the outcome of physical access attempts.
- **Weekend activity:** Derived from the event timestamp.
- **Temporal proximity:** Calculated using the time difference between potentially related physical and cyber events.

3.3 Ground Truth Labelling

To support supervised machine learning classification and evaluate the proposed hybrid reasoning framework, ground-truth risk labels are assigned to the generated cyber-physical security scenarios. Each scenario is assessed based on physical access activities, cybersecurity indicators, user roles, normal working hours, and temporal relationships between events. A predefined risk assessment rubric is used to assign scores to relevant security indicators. The scores are combined to determine the overall risk category. The scoring criteria and thresholds are established before model training and documented

to ensure consistency and reproducibility. Figure 1 illustrates the distribution of the three ground-truth risk categories: Low, Medium, and High, across the labelled security scenarios

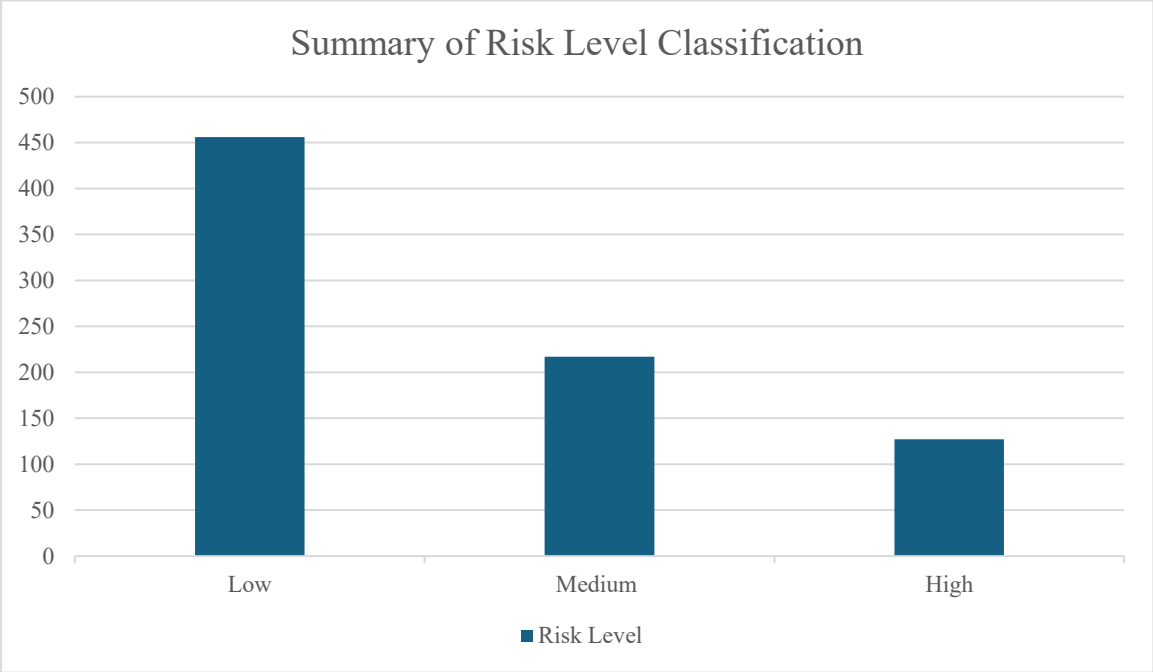


Figure 1 - Distribution of Risk Classification

Where each risk defined as the following:

- **Low Risk:** Activities consistent with expected user behaviour, normal physical access, and minimal or no suspicious cybersecurity indicators. minor deviations may be present without significant security concerns.
- **Medium Risk:** Noticeable suspicious activity in either the cyber or physical domain, or a combination of moderate indicators that may require further investigation.
- **High Risk:** Strong indicators of potential security threats, particularly when suspicious cyber and physical activities are associated with the same user and occur within a relevant time window.

To clarify how each level is assigned to a given scenario, the following table shows examples of the evidence used to determine these labels:

Table 4 - Risk Level Scenarios

Scenario id	Risk level	Description
S061	LOW	Authorized access to a restricted location is consistent with the user's organizational role, while cybersecurity activity shows no significant anomalies.
S003	MEDIUM	Physical access occurs outside normal working hours, but the associated cybersecurity activity shows limited suspicious indicators.
S005	HIGH	Access to a restricted location outside normal working hours is followed by suspicious cybersecurity activity, including repeated failed logins and sensitive-file access, within a short time interval.

4 Knowledge Graph/System Design

4.1 Rationale and Knowledge Representation

The Knowledge Graph (KG) provides the knowledge-representation layer of the hybrid cyber-physical security framework. Although the datasets can be combined into a single table, the graph explicitly represents how users, assigned devices, physical-access events, locations, and cybersecurity events relate to one another. This supports traceable multi-hop retrieval of a correlated scenario and makes the event relationships available to the symbolic reasoning component.

The KG is not itself a risk classifier. It supplies connected facts, checks the consistency of user-device-event relationships, and computes temporal context. The rule-based component subsequently applies the agreed security conditions to produce a symbolic risk assessment and priority.

4.2 Property-Graph Schema

Neo4j was selected to implement a property graph with five node labels and five directed relationship types.

Node label	**Identifier**		**Relevant properties**
User	user_id		role, department, normal_start, normal_end
Device	device_id		device_id
PhysicalEvent	event_id		timestamp, access_result
Location	location		location_type
CyberEvent	event_id		timestamp, severity, failed_logins, files_accessed, sensitive_files, usb_connected
Relationship	**Direction**		**Meaning**
USES	User → Device		Assigned device
PERFORMED	User → PhysicalEvent		Physical-access activity
AT	PhysicalEvent → Location		Event location
GENERATED	Device → CyberEvent		Device-generated cyber activity
CORRELATED_WITH	PhysicalEvent → CyberEvent		Intended paired events in one scenario

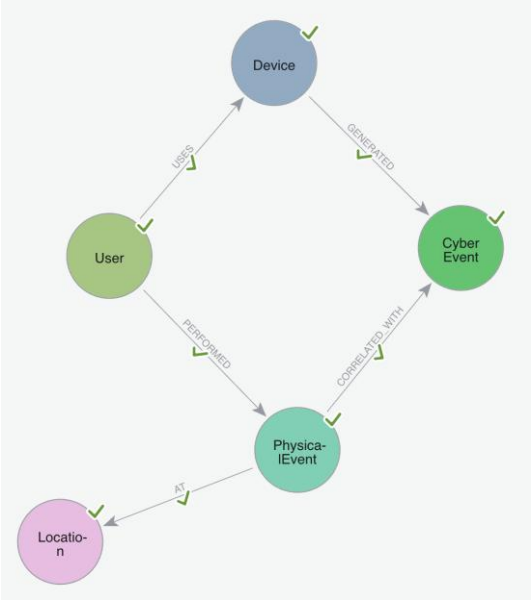


Figure 2. Knowledge Graph schema and relationship design.

4.3 Graph Construction and Scenario Correlation

The Users, Physical Events, and Cyber Events datasets were imported into Neo4j and mapped to the graph schema using stable identifiers. Locations are represented as shared nodes rather than repeated for every access event. The 800 intended physical-cyber scenario pairs are represented by explicit CORRELATED_WITH relationships. This prevents temporal proximity alone from creating unintended event pairings when several events occur near one another.

A complete scenario is retrieved by following User–PERFORMED–PhysicalEvent–AT–Location and User–USES–Device–GENERATED–CyberEvent, with the PhysicalEvent–CORRELATED_WITH–CyberEvent relationship connecting the two domains. These traversals retrieve the associated context without assigning a security risk.

Node type	**Count**
User	50
Device	50
PhysicalEvent	800
CyberEvent	800
Location	9
Total	1,709
Relationship type	**Count**
USES	50
PERFORMED	800
AT	800
GENERATED	800
CORRELATED_WITH	800
Total	3,250

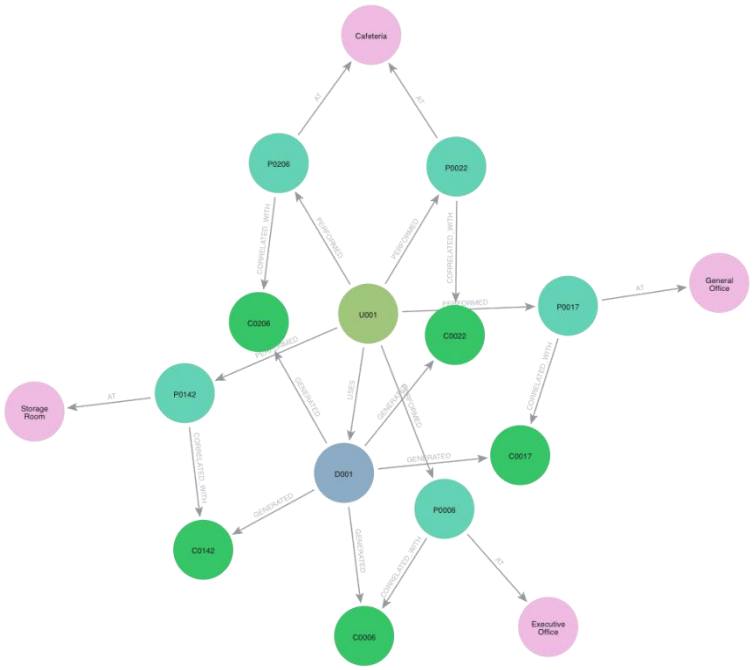


Figure 3. Implemented Knowledge Graph showing connected physical and cyber activities.

4.4 Context Retrieval and Temporal Features

Cypher queries retrieve stored attributes and derive temporal context from the graph. The after_hours indicator compares each physical-event timestamp with the corresponding user's normal_start and normal_end values. The time_gap_minutes value is independently calculated from the correlated physical and cyber timestamps. By contrast, location_type and access_result are existing dataset attributes retrieved through the graph, not newly inferred facts.

The original timestamp import required correction to preserve the intended local clock times. Following conversion to local datetime values, the after-hours calculation produced 252 after-hours and 548 within-hours events. The calculated physical-to-cyber gaps were also checked against the stored correlation values and matched within rounding tolerance.

The finalized temporal-proximity threshold used by the subsequent symbolic reasoning is 10 minutes, inclusive. This threshold is a reasoning condition; it does not alter the underlying event timestamps or the graph's time-gap calculation.

4.5 Representative Correlated Scenario

A representative scenario demonstrates the graph's two complementary forms of evidence. The graph view shows the connected user, assigned device, physical event, location and cyber event. The table view shows the same scenario's stored facts and calculated context, including after-hours status and temporal gap. Both figures should use identical physical and cyber event identifiers.

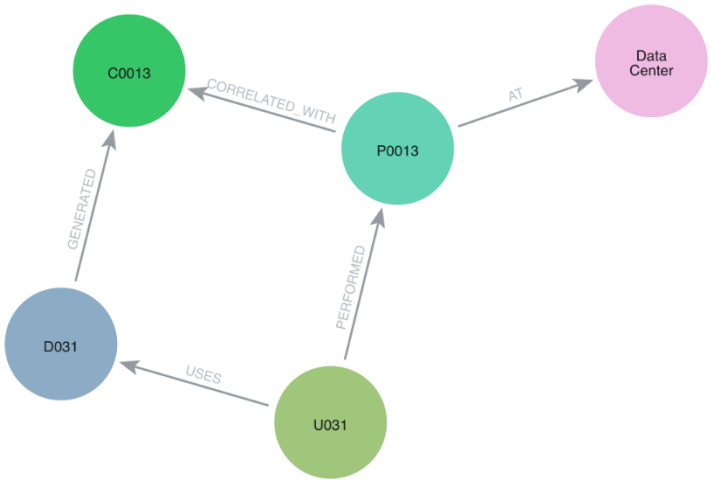


Figure 4. Graph representation of one correlated cyber-physical scenario.

```
7cb230c8$ MATCH (u:User)-[:PERFORMED]->(p:PhysicalEvent)-[:AT]->(l:Location) MATCH (p)-[:CORRELATED_WITH]->(c:CyberEvent) MATCH (u)-[:USES]->(d:Device)-[:
```

Run

Table | Raw | Search table... | Graph chip

user	device	physical_eve	physical_	location	location_ty	access_resu	after_hour	cyber_eve	cyber_time	cyber_severit	failed_logi	files_acces	sensitive_file	usb_connecte	time_gap_minute
"U031"	"D031"	"P0013"	2026-04-02T01:38:02	"Data Center"	"Restricted"	"Granted"	TRUE	"C0013"	2026-04-02T02:02:55	"Medium"	2	36	5	FALSE	24.88

Figure 5. Retrieved physical, cyber and temporal context for the same scenario.

4.6 Graph Validation

Validation queries checked graph population, relationship completeness, and scenario pairing. The implemented graph contains 1,709 nodes and 3,250 relationships. No physical event was missing its User or Location link, and no cyber event was missing its Device link. Each physical event had exactly one outgoing CORRELATED_WITH relationship to a cyber event.

Validation measure	**Result**
Physical events without User	0
Physical events without Location	0
Cyber events without Device	0
Correlations per physical event	Minimum 1; maximum 1
After-hours events	252
Within-hours events	548
Time-gap comparison	Matched stored values within rounding tolerance

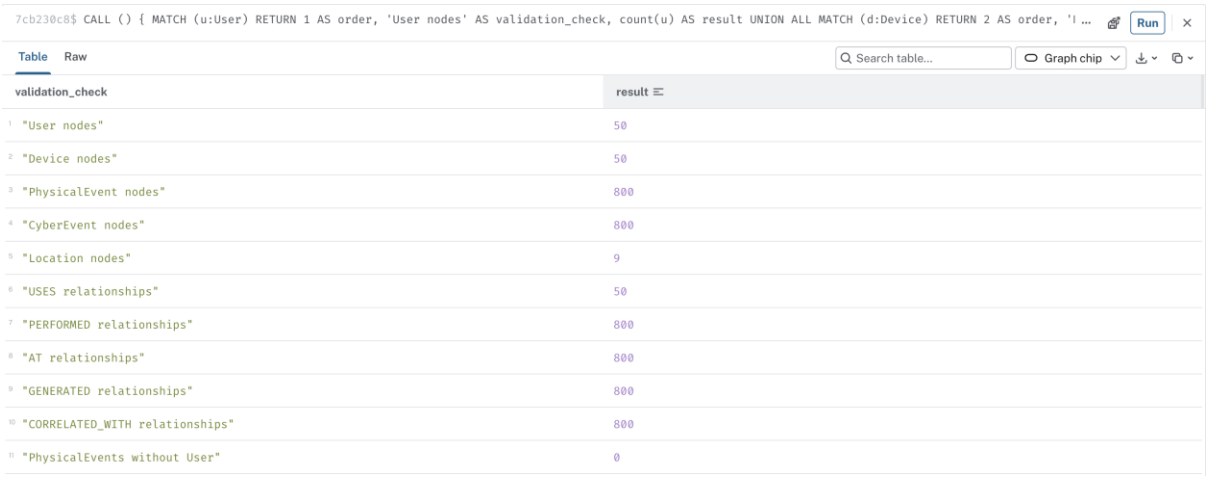


Figure 6. Knowledge Graph structural and temporal validation.

4.7 Integration with Symbolic Reasoning

To integrate the graph with the symbolic reasoning implementation, a Cypher extraction query exports one contextual record for each of the 800 scenarios. The kg_scenario_context.csv output supplies the current physical and cyber facts, calculated after-hours status, and physical-to-cyber time gap. A separate kg_relationships.csv output supplies scenario-aligned user/device relationship checks and supporting graph-path evidence. The Python symbolic reasoning notebook integrates the KG-extracted scenario context and relationship-validation outputs to evaluate the predefined security rules and infer scenario-level risk classifications.

5 Machine Learning Methodology and Evaluation

This section presents the machine learning methodology and evaluation results for cyber-physical risk classification. Different classification models, including Decision Tree, Random Forest and Gradient Boosting, are evaluated using cyber-only, physical-only and hybrid feature sets to examine whether combining both domains improves risk prediction. Each model classifies scenarios as Low, Medium or High risk, using the same test scenarios and reference labels to ensure a fair comparison.

5.1 Machine Learning Methodology

5.1.1 Classification Approach and Model Selection

The risk prediction task is formulated as supervised multiclass classification, with Low, Medium and High representing the three risk categories. Each scenario retains the same reference label across the cyber-only, physical-only and hybrid experiments. Only the input features change, allowing the effect of combining cyber and physical information to be assessed consistently.

Decision Tree serves as the baseline because its predictions can be traced through explicit rules and feature thresholds. This makes the classification process easier for security analysts to examine and provides a reference for evaluating the other models. Random Forest and Gradient Boosting are included to investigate whether using multiple decision trees improves predictive performance. Random Forest combines predictions from several trees, while Gradient Boosting builds trees sequentially to reduce prediction errors. The comparison considers both classification performance and the ease of explaining model decisions. Each classifier is evaluated using cyber-only, physical-only and hybrid feature sets, resulting in nine experiments.

5.1.2 Feature Selection

Feature selection focuses on security activity and contextual information relevant to predicting the overall risk of each scenario. The cyber-only baseline includes seven predictors: failed_logins, files_accessed, sensitive_files, usb_connected, cyber_after_hours, cyber_hour and cyber_is_weekend. These describe cyber activity and when it occurs. The physical-only baseline includes five predictors: restricted_location, access_denied, physical_after_hours, physical_hour and physical_is_weekend. These describe physical access conditions and their timing.

The hybrid model combines these twelve predictors with time_gap_minutes, producing thirteen input features for each merged scenario. This allows cyber and physical evidence to be considered together when predicting overall risk. The time gap is included only in the hybrid model because it measures the interval between the linked physical and cyber events.

Identifiers and raw timestamps are excluded from model inputs, while relevant timing information is represented through the selected time features. risk_score is excluded to prevent direct use of the score associated with the target label. severity and severity_numeric are also excluded from all experiments.

5.1.3 Training and Model Tuning

The dataset is divided into 80% training data and 20% testing data, resulting in 640 training scenarios and 160 test scenarios. A stratified split preserves the approximate proportion of each risk category, while random_state=42 makes the split reproducible. The test set contains 91 Low-risk, 43 Medium-risk and 26 High-risk scenarios. All nine experiments use the same training and testing records to ensure a fair comparison.

Model settings are selected through five-fold stratified cross-validation within the training data. The training set is divided into five subsets, with each subset used once for validation while the remaining four are used for training. GridSearchCV compares the candidate settings using the average macro F1 score across these folds.

The settings examined differ by classifier. For Decision Tree, these include tree depth, minimum samples per leaf and class weighting. Random Forest uses 150 trees and varies tree depth, minimum samples per leaf, feature sampling and class weighting. Gradient Boosting varies the number of trees, learning rate and tree depth. Each model is then trained on the full training set using the selected settings. Cross-validation performance determines the recommended hybrid classifier, while the separate test set provides the final evaluation results.

5.2 Evaluation Methodology

The evaluation examines whether combining cyber and physical information improves risk classification compared with using either domain alone. Decision Tree, Random Forest and Gradient Boosting are assessed across the three feature sets using the same 160 test scenarios. Performance is compared using accuracy, macro precision, macro recall and macro F1, with High-risk recall reported separately to assess the detection of High-risk scenarios.

Confusion matrices and misclassified scenarios are examined to identify the types of errors made by each model, particularly cases where risk is underestimated. Feature importance is also analysed to assess the contribution of cyber and physical indicators to the selected hybrid model. These analyses support the comparisons and interpretation presented in the following section.

5.3 Machine Learning Results

5.3.1 Overall Performance

Table 5 presents the test performance of Decision Tree (DT), Random Forest (RF) and Gradient Boosting (GB) across the three feature sets. Precision, recall and F1 are macro-averaged, and all results are expressed as percentages.

Table 5. Test performance across feature sets and classifiers.

Feature set	Model	Accuracy	Precision	Recall	F1	High recall
Cyber-only	DT	88.75	86.38	85.87	85.80	76.92
Cyber-only	RF	96.25	95.92	96.89	96.23	96.15
Cyber-only	GB	95.62	95.63	95.70	95.61	96.15
Physical-only	DT	69.38	57.41	56.96	56.33	46.15
Physical-only	RF	71.25	59.99	56.44	57.27	30.77
Physical-only	GB	72.50	63.08	58.19	59.32	38.46
Hybrid	DT	93.12	93.48	90.67	91.83	84.62
Hybrid	RF	98.12	98.11	98.49	98.30	100.00
Hybrid	GB	96.88	96.73	96.84	96.73	96.15

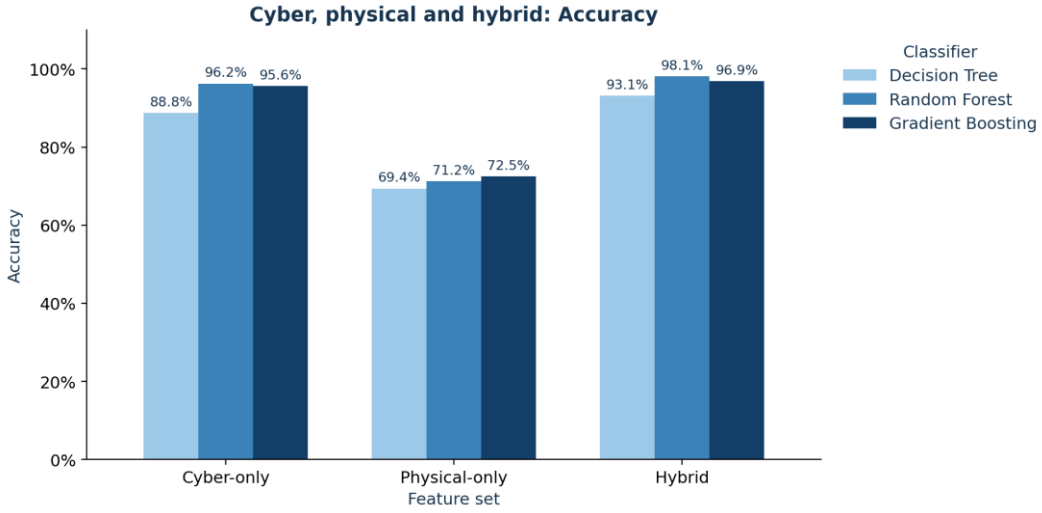


Figure 7. Accuracy across cyber-only physical-only and hybrid feature sets.

Figure 7 shows that the hybrid feature set achieves the highest accuracy for all three classifiers: 93.12% for Decision Tree, 98.12% for Random Forest and 96.88% for Gradient Boosting. Compared with cyber-only features, these represent improvements of 4.38, 1.88 and 1.25 percentage points, respectively. Physical-only accuracy ranges from 69.38% to 72.50%. These results indicate that cyber activity provides strong predictive information, while physical context further improves overall risk classification.

Hybrid Random Forest correctly classifies 157 of 160 test scenarios and identifies all 26 High-risk cases. Its selection is based on the highest training cross-validation macro F1 of 93.80%, compared with 93.14% for hybrid Gradient Boosting and 90.55% for hybrid Decision Tree. The test results support this selection, with the three misclassifications occurring between Low and Medium risk.

5.3.2 Cyber-only and Physical-only Performance

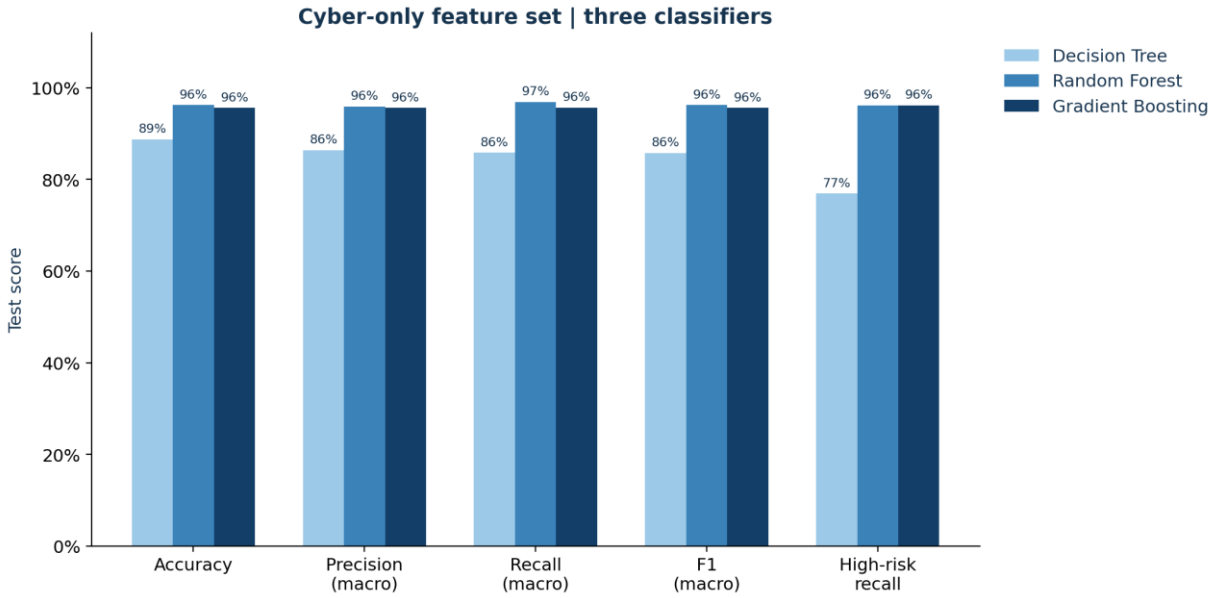


Figure 8. Cyber-only test metrics for the three classifiers.

Figure 8 compares the three classifiers using cyber-only features. Random Forest achieves the highest accuracy at 96.25%, followed by Gradient Boosting at 95.62% and Decision Tree at 88.75%. Random Forest and Gradient Boosting each identify 25 of the 26 High-risk scenarios, achieving 96.15% High-risk recall. Decision Tree identifies 20 High-risk scenarios, resulting in 76.92% recall. These results show that Random Forest and Gradient Boosting provide stronger overall classification and High-risk detection than Decision Tree when using cyber information alone.

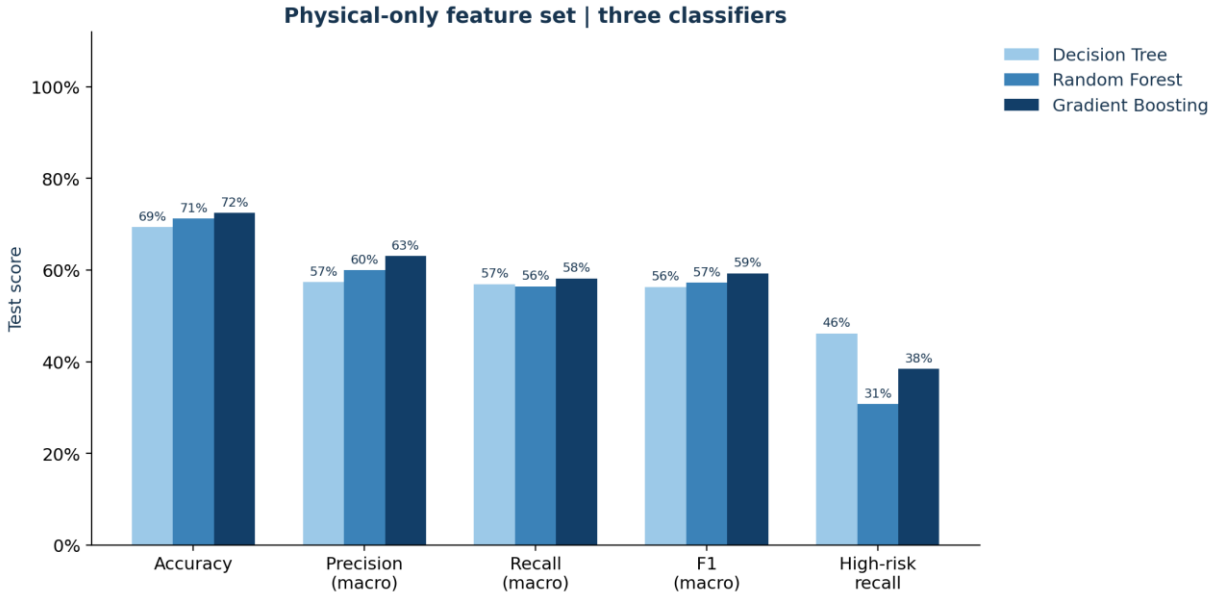


Figure 9. Physical-only test metrics for the three classifiers.

Figure 9 compares the classifiers using physical-only features. Gradient Boosting achieves the highest accuracy at 72.50%, followed by Random Forest at 71.25% and Decision Tree at 69.38%. Gradient Boosting also records the highest macro F1 at 59.32%. However, Decision Tree achieves the strongest High-risk recall at 46.15%, identifying 12 of the 26 High-risk scenarios, compared with eight for Random Forest and ten for Gradient Boosting. These results show that higher overall accuracy does not

necessarily correspond to better High-risk detection. All three models miss more than half of the High-risk cases, indicating the limitations of physical-access information alone for predicting overall scenario risk.

5.3.3 Hybrid Performance

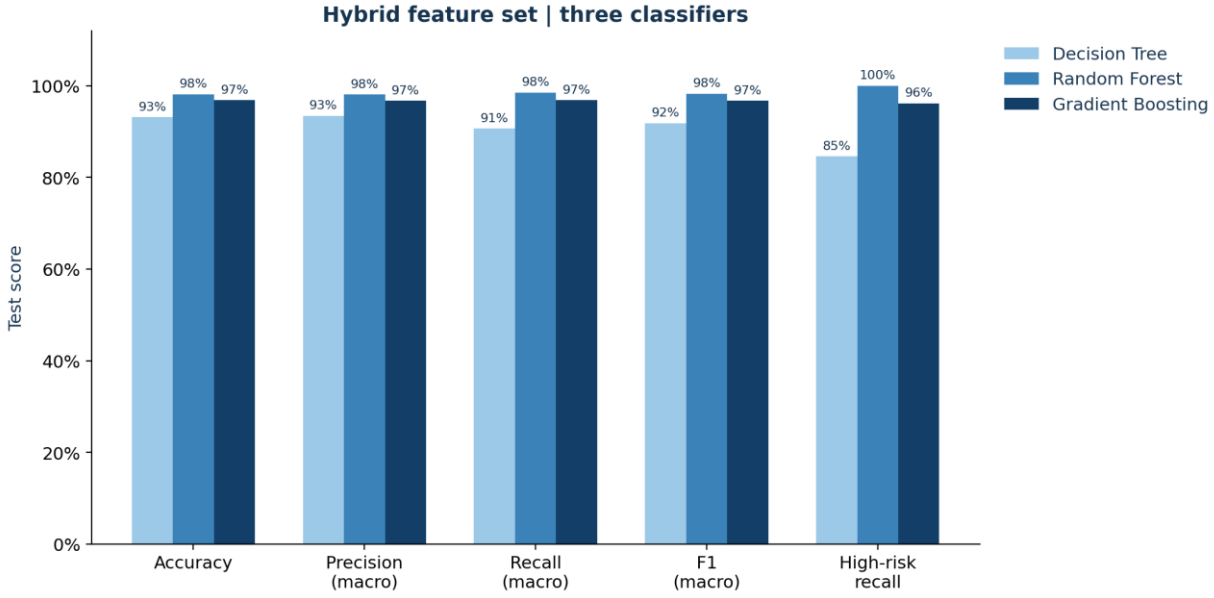


Figure 10. Hybrid test metrics for the three classifiers.

Figure 10 compares the classifiers using combined cyber and physical features. Random Forest achieves the highest accuracy at 98.12%, followed by Gradient Boosting at 96.88% and Decision Tree at 93.12%. Random Forest also records the highest macro F1 at 98.30% and correctly identifies all 26 High-risk test scenarios. Gradient Boosting detects 25 High-risk cases, while Decision Tree detects 22, corresponding to recall scores of 96.15% and 84.62%, respectively.

Random Forest is selected based on its training cross-validation macro F1 of 93.80%, slightly above Gradient Boosting at 93.14%. The test results support the selection of Random Forest, although its cross-validation performance is only slightly higher than that of Gradient Boosting. The model correctly identifies all 26 High-risk scenarios in the test set.

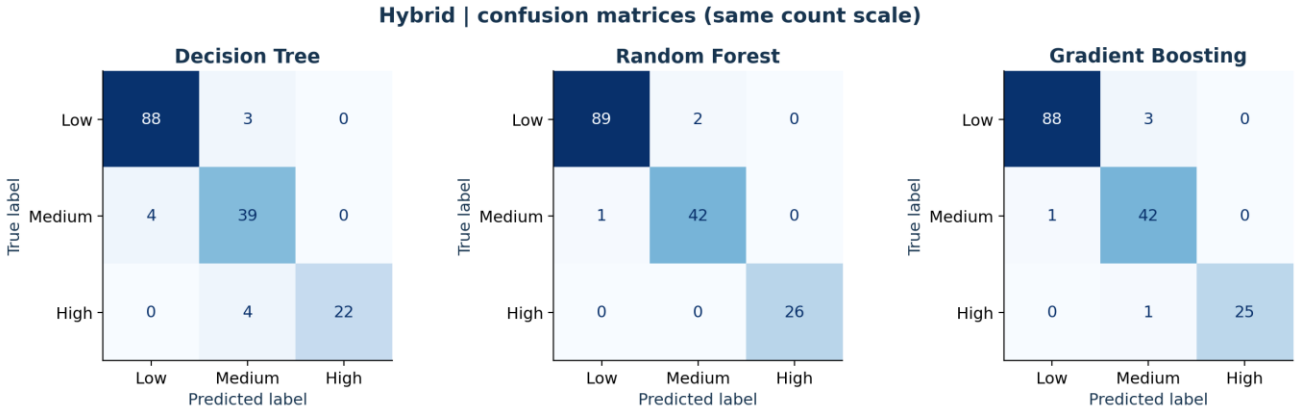


Figure 11. Hybrid confusion matrices with actual risk in rows and predicted risk in columns.

Figure 11 shows the classification errors for the three hybrid models. The selected Random Forest correctly classifies 89 of 91 Low-risk scenarios, 42 of 43 Medium-risk scenarios and all 26 High-risk scenarios. Its three errors consist of two Low-risk scenarios predicted as Medium and one Medium-risk

scenario predicted as Low. No High-risk scenarios are missed, and no Low- or Medium-risk scenarios are incorrectly classified as High.

5.3.4 Features Contribution and Classification Errors

Table 6. Leading feature importances for hybrid Random Forest.

Feature	Importance
failed_logins	21.4%
sensitive_files	16.3%
files_accessed	12.2%
restricted_location	10.7%
usb_connected	7.2%
physical_after_hours	6.9%
cyber_after_hours	5.7%
access_denied	5.1%

Table 6 shows that failed_logins has the highest importance at 21.4%, followed by sensitive_files at 16.3% and files_accessed at 12.2%. restricted_location is the strongest physical predictor at 10.7%, with additional contributions from physical after-hours access and denied access. These findings indicate that the selected hybrid model draws on both cyber activity and physical-access context. The importance values describe the relative contribution of features to the fitted model; they do not explain how a particular feature affects an individual prediction.

Table 7. Misclassifications by the selected hybrid Random Forest.

Actual risk	Predicted risk	Cases
Low	Medium	2
Medium	Low	1

Table 7 presents the three misclassifications made by hybrid Random Forest across the 160 test scenarios. Two Low-risk scenarios are classified as Medium, while one Medium-risk scenario is classified as Low. The errors are limited to adjacent risk categories, with no High-risk scenarios misclassified. This indicates that the remaining classification difficulty lies in distinguishing Low from Medium risk within the evaluated test set.

5.3.5 Summary of Machine Learning Findings

Overall, the results support the use of combined cyber and physical features for risk classification, with hybrid models achieving higher accuracy than either domain alone. Hybrid Random Forest is selected based on training cross-validation and achieves 98.12% test accuracy, correctly classifying 157 of 160 scenarios. These findings demonstrate the benefit of combining both domains within the evaluated proof of concept.

6 Symbolic Reasoning Methodology

6.1 Overview

The symbolic reasoning component was developed to provide a transparent and interpretable mechanism for assessing cyber-physical security risks. While the machine learning component predicts risk levels based on patterns learned from the dataset, symbolic reasoning applies explicitly defined security rules to identify suspicious activities and explain the evidence supporting each assessment.

The implementation builds upon the integrated cyber-physical dataset and the Knowledge Graph (KG) developed in the preceding stages. The KG provides contextual relationships between users, devices, physical access events, and cybersecurity events, enabling the reasoning component to identify suspicious activities occurring across both domains.

The symbolic reasoning process consists of four main activities: validating KG relationships, evaluating security indicators against predefined rules, determining the overall symbolic risk level, and recording the evidence supporting the assessment.

6.2 Knowledge Graph Integration and Validation

Before applying the symbolic rules, the framework integrates the KG outputs with the 800 paired cyber-physical scenarios.

Two datasets are utilized:

- kg_relationships.csv, which provides relationship evidence and confirms whether the physical and cybersecurity events are associated with the same user and device.
- kg_scenario_context.csv, which provides contextual information, including event identifiers, timestamps, locations, access outcomes, and cybersecurity indicators.

The integration process verifies that each scenario corresponds to the correct physical and cyber events. Rather than relying solely on the ordering of records, the framework matches the physical and cybersecurity event identifiers and validates the consistency of the associated user, device, location, and event attributes. These checks are performed before executing the symbolic rules to ensure that the reasoning process is based on consistent contextual information.

This step is particularly important for detecting cross-domain risks, where a suspicious physical event should only be associated with a cybersecurity event when the relationship between the events is supported by the KG.

6.3 Symbolic Rule Development

A total of ***11 symbolic rules*** were implemented to identify suspicious physical activities, cybersecurity anomalies, and combined cyber-physical risks.

The rules were designed around the security indicators available in the dataset, including access outside working hours, restricted-location access, failed authentication attempts, sensitive-file access, USB connectivity, and event severity.

The rules are categorized into three groups.

Table 8. Symbolic Reasoning Rules.

Rule	Category	Rule condition	Inference
R1	Physical	Access outside working hours AND restricted location	Physical anomaly

R2	Physical	Access denied AND restricted location	Physical anomaly
R8	Physical	Access outside working hours AND access denied	Physical anomaly
R3	Cyber	Failed logins ≥ 5 AND sensitive files accessed ≥ 10	Cyber anomaly
R4	Cyber	USB connected AND sensitive files accessed ≥ 10	Cyber anomaly
R5	Cyber	High cybersecurity severity AND failed logins ≥ 5	Cyber anomaly
R9	Cyber	High cybersecurity severity AND sensitive files accessed ≥ 10	Cyber anomaly
R10	Cyber	High cybersecurity severity AND USB connected	Cyber anomaly
R11	Cyber	USB connected AND failed logins ≥ 5	Cyber anomaly
R7	Cross-domain	Physical anomaly AND cyber anomaly AND KG-confirmed same user	Elevated cyber-physical risk
R6	Cross-domain	R7 conditions AND time difference ≤ 10 minutes	Time-critical cyber-physical risk

The first nine rules identify suspicious activities within the individual security domains. Rules R6 and R7 extend the analysis by examining whether physical and cybersecurity anomalies are associated with the same user.

R7 identifies an elevated cyber-physical risk when both domains exhibit anomalies and the KG confirms a common user. R6 introduces an additional temporal condition, requiring the associated events to occur within ten minutes of each other.

The ten-minute threshold was aligned with the updated temporal proximity criterion adopted during dataset preparation.

6.4 Symbolic Rule Classification

Following rule execution, the framework assigns one of three risk levels to each scenario.

Table 9. Symbolic risk classification.

Risk Level	Classification Logic
Low	No configured physical or cybersecurity anomaly rules are triggered
Medium	At least one physical or cybersecurity anomaly is identified, without a KG-confirmed combination of both domains
High	Physical and cybersecurity anomalies are identified and associated with the same user through the KG

In addition to the risk classification, the framework assigns a priority level to support the identification of cases requiring attention.

A scenario is classified as ***Time-critical*** when R6 is triggered, Elevated when R7 is triggered without R6, and Normal otherwise.

The priority classification is maintained separately from the risk level. For example, a High-risk scenario may be assigned either Elevated or Time-critical priority depending on whether the temporal proximity condition is satisfied.

6.5 Evidence Collection and Explainability

To support transparent reasoning, each triggered rule generates a structured evidence record containing the rule identifier, the inferred anomaly, and the factual indicators that satisfied its conditions.

For example, when R3 is triggered, the evidence record includes the number of failed login attempts and the number of sensitive files accessed.

For cross-domain rules, the evidence additionally identifies the relationship between the physical and cybersecurity events. R7 records the KG-confirmed same-user relationship, while R6 records the time difference and the applicable ten-minute threshold.

The framework retains the complete list of triggered rules and supporting evidence for each scenario. These outputs are subsequently used by the hybrid reasoning component and the Large Language Model (LLM) to explain the final risk assessment.

Importantly, the symbolic reasoning engine does not use the reference risk labels or risk scores as inputs when determining its predictions. These values are reserved for the evaluation stage.

6.6 Symbolic Reasoning Results

The symbolic reasoning engine was executed across all 800 cyber-physical scenarios, generating a symbolic risk classification, priority level, triggered-rule list, and supporting evidence for each scenario.

For comparative evaluation, symbolic predictions were assessed against the reference labels for the same 160 held-out scenarios used to evaluate the machine learning and hybrid approaches.

Table 10. Symbolic reasoning performance.

Evaluation Metric	Classification logic
Accuarcy	96.25%
Macro F1-score	94.83%
High-risk recall	96.15%
Low/Medium scenarios incorrectly classified as High	3

The results indicate that the symbolic reasoning component achieved high agreement with the reference risk classifications while maintaining an interpretable decision process.

The High-risk recall of 96.15% demonstrates that the rules identified most of the scenarios labelled as High risk. However, three scenarios with Low or Medium reference labels were classified as High risk, illustrating the potential for conservative rule-based reasoning to produce false escalations.

These findings should be interpreted in the context of the dataset, where reference labels were generated using a predefined scoring rubric. Consequently, the evaluation measures agreement with the rubric-derived classifications rather than independently verified security incidents.

7 Hybrid Reasoning Methodology

7.1 Overview

The hybrid reasoning component was developed to combine the predictive capabilities of machine learning with the transparency and contextual reasoning provided by symbolic rules.

The machine learning component uses a **Random Forest classifier** to predict security risk levels from the integrated cyber-physical features. In parallel, the symbolic reasoning component independently evaluates each scenario using the predefined security rules and KG-supported relationships.

The outputs of both components are subsequently combined to produce a final hybrid risk assessment. This approach allows the framework to consider both data-driven predictions and explicitly defined security conditions, while preserving the reasoning evidence associated with each decision.

7.2 Integration of Machine Learning and Symbolic Predictions

The integration was performed using the 160 held-out scenarios from the machine learning evaluation dataset.

For each scenario, the framework retrieves the Random Forest prediction and matches it with the corresponding symbolic reasoning output using the unique scenario identifier. The resulting integrated record contains the machine learning prediction, symbolic risk classification, triggered rules, supporting evidence, KG relationship information, and the final hybrid assessment.

This structure enables direct comparison between the two reasoning approaches and provides the information required to investigate cases where their predictions differ.

7.3 Hybrid Decision Mechanism

A conservative risk-selection policy was adopted to determine the final hybrid risk level. Under this policy, the framework selects the higher risk classification between the Random Forest prediction and the symbolic reasoning result, as illustrated below.

The decision mechanism is defined as:

$$R_{\text{Hybrid}} = \max\left(R_{\text{ML}}, R_{\text{Symbolic}}\right)$$

Table 11. Examples of hybrid risk classification.

ML Prediction	Symbolic Risk	Final Hybrid Risk
Low	Low	Low
Low	**Medium**	Medium
Medium	**High**	High
High	Medium	High

This policy was selected to prioritize potentially significant security risks when the two approaches produce different classifications.

For example, when the machine learning model predicts Medium risk but the symbolic rules identify linked physical and cybersecurity anomalies indicating High risk, the final hybrid assessment is classified as High.

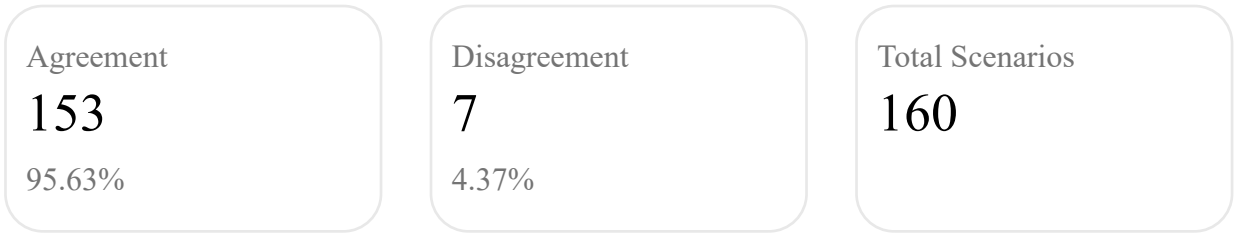
Conversely, if the machine learning model predicts High risk while symbolic reasoning indicates Medium-risk, the framework retains the High-risk classification.

Although this policy supports conservative risk prioritization, it does not necessarily improve overall classification accuracy. Selecting the higher risk may also increase false-positive escalations. Therefore, its effectiveness was assessed through comparative evaluation.

7.4 Agreement and Disagreement Analysis

To evaluate the consistency between the two reasoning approaches, an agreement indicator was introduced. A scenario is classified as Agree when the ML and symbolic reasoning components predict the same risk level, and Disagree when their predictions differ.

The final integration results demonstrated a high level of consistency between the two approaches.



The seven disagreement cases illustrate how the hybrid policy resolves differences between the machine learning and symbolic predictions. In each case, the higher classification was retained as the final assessment. None of the seven cases involved a direct Low-versus-High disagreement.

These cases also demonstrate the value of preserving the individual component predictions rather than reporting only the final risk level. This allows analysts to identify differences between the two methods and examine the supporting evidence before determining whether further investigation is necessary.

7.5 Comparative Performance Evaluation

The Random Forest, symbolic reasoning, and hybrid reasoning approaches were evaluated using the same 160 held-out scenarios and their corresponding reference risk labels.

Four evaluation measures were considered: accuracy, Macro F1-score, High-risk recall, and the number of Low- or Medium-risk scenarios incorrectly classified as High risk.

Table 12. Comparative evaluation results.

Evaluation Metric	Rnadom Forest	Symbolic	Hybrid
Accuracy	98.13%	96.25%	96.25%
Macro F1-score	98.30%	94.83%	95.30%
High-risk recall	100%	96.15%	100%
Low/Medium escalated to High	0	3	3

The Random Forest model achieved the highest overall classification accuracy and Macro F1-score among the three approaches.

The symbolic reasoning component also demonstrated strong performance, achieving 96.25% accuracy while providing explicit rules and supporting evidence for its predictions.

The hybrid approach achieved 96.25% accuracy and a Macro F1-score of 95.30%. Although its overall accuracy was lower than the Random Forest model, it maintained 100% High-risk recall, meaning that all High-risk scenarios in the held-out test set were identified. The hybrid approach also produced three incorrect High-risk escalations, compared with none for the Random Forest model.

These findings demonstrate the trade-off associated with the selected fusion policy. While the hybrid framework preserves the higher risk classification and supports explainable decision-making, it does not outperform the Random Forest model in overall classification accuracy. Therefore, the primary contribution of the hybrid approach lies in combining machine learning predictions with explicit symbolic reasoning, KG-supported evidence, and transparent risk prioritization rather than improving classification accuracy alone.

8 LLM Methodology

8.1 Overview

A Large Language Model (LLM) was integrated into the framework to generate human-readable explanations of the final hybrid risk assessments. The purpose of this component is to improve the interpretability of the system's outputs by translating structured reasoning results into concise explanations that security analysts can understand.

Unlike the machine learning and symbolic reasoning components, the LLM does not independently classify scenarios or determine their risk levels. Instead, it receives the final risk assessment and the evidence generated by the preceding stages and produces a textual explanation of the decision. This separation ensures that the LLM functions as an explanation mechanism rather than an additional risk classifier.

8.2 LLM Input Preparation

A structured input payload was developed to provide the LLM with the information required to explain each assessment. The payload contains the final hybrid risk, individual component predictions, triggered symbolic rules, supporting evidence, KG relationship indicators, and relevant contextual information.

Table 13. LLM input structure.

Input	Description
Final hybrid risk	Final Low, Medium, or High classification
ML prediction	Risk predicted by the Random Forest model
Symbolic risk	Risk derived from symbolic reasoning
Triggered rules	List of activated symbolic rules
Supporting evidence	Indicators and conditions supporting each rule
KG same-user indicator	Whether the events are associated with the same user
KG same-device indicator	Whether the events are associated with the same device
Location	Physical location associated with the scenario
Time difference	Temporal separation between the physical and cyber events
Agreement status	Whether ML and symbolic predictions agree

Priority	Normal, Elevated, or Time-critical

The input was intentionally restricted to the evidence relevant to the assessment. The reference risk score and reference risk label were excluded to prevent the LLM from relying on evaluation information that was not available to the reasoning components. Additionally, raw user and device identifiers were also excluded from the explanation payload, reducing unnecessary disclosure of identifying information.

8.3 Explanation Generation

The LLM component was implemented using the OpenAI API with the gpt-4o-mini model. For each scenario, the structured input payload was submitted to the LLM together with instructions governing the explanation. The instructions required the LLM to preserve the supplied final risk level, describe the triggered symbolic rules and their supporting evidence, and distinguish observed indicators from inferred security risks.

When the machine learning and symbolic reasoning components disagreed, the LLM was instructed to identify both predictions and explain that the final classification was determined by selecting the higher risk level. The model was also instructed not to invent additional events, causes, or supporting facts and to produce concise, professional explanations.

The explanation-generation process was executed across all 160 held-out hybrid scenarios, producing one explanation for each final risk assessment.

8.4 LLM Explanation Evaluation Process & Results

To evaluate the quality of the generated explanations, a manual review was conducted using a reproducible random sample of 30 explanations selected from the 160 generated outputs.

Four evaluation criteria were established.

Table 14. LLM explanation evaluation criteria.

Input	Description
Risk consistency	Determine whether the explanation preserves the final hybrid risk level
Evidence faithfulness	Determine whether the explanation accurately represents the triggered rules and supporting evidence
No hallucination	Determine whether the explanation avoids unsupported factual claims
Explanation clarity	Determine whether the explanation communicates the assessment clearly and logically

Each explanation was manually assessed against the four criteria using Pass/Fail scale. The results were then aggregated to calculate the pass rate for each criterion for 30 randomly selected explanations, as illustrated below.

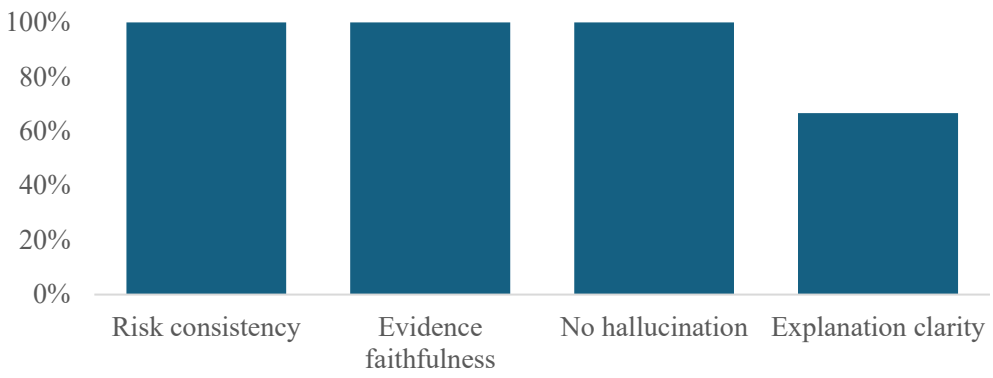


Figure 12. LLM explanation evaluation results.

The manual evaluation results indicate that all 30 reviewed explanations preserved the final hybrid risk classification and accurately reflected the supporting evidence according to the review criteria. No unsupported factual claims were identified in the reviewed sample.

However, explanation clarity achieved a lower pass rate of 66.67%, with ten explanations failing to meet the established clarity criterion. This finding indicates that, although the explanations were generally consistent with the supplied assessment and evidence, some were insufficiently clear in communicating a clear explanation to the reader.

The results highlight the distinction between factual faithfulness and explanation quality. An explanation may accurately reflect the available evidence while still requiring improvements in structure, specificity, or readability.

In conclusion, the LLM component demonstrates the feasibility of producing natural-language explanations from structured hybrid reasoning outputs. Explanations are grounded in predicted risk levels, triggered rules and supporting evidence, keeping the LLM's role focused on explaining existing outputs rather than assigning new risk classifications.

Manual evaluation identified clarity as an area for improvement. Refining the instructions could improve the organisation and readability of the explanations. Although explanations were generated for the full test set, the reported quality scores are based on a manually reviewed sample of 30 randomly selected scenarios.

Future evaluation could include multiple independent reviewers and a more detailed scoring rubric to improve consistency and assess the usefulness of the explanations for security investigations.

9 Cyber-Physical Security Intelligence Dashboard

As the final deliverable, a web-based ***Cyber-Physical Security Intelligence (CPSI) Dashboard*** was developed to demonstrate the practical application of the proposed hybrid reasoning framework.

The dashboard integrates the outputs of the Random Forest model, symbolic reasoning, Knowledge Graph, and LLM-generated explanations into a unified interface, allowing users to monitor security risks, investigate scenarios, and examine supporting evidence.

The dashboard consists of five main sections:

- **Security Overview:** Summarizes risk classifications, priorities, and triggered rules.
- **High-Risk Events:** Supports investigation of high-risk and time-critical scenarios.

- **Disagreement Review:** Highlights differences between ML and symbolic predictions for analyst review.
- **Knowledge Graph Explorer:** Visualizes relationships between users, devices, locations, and security events.
- **Analytics & Evaluation:** Presents model performance and explanation evaluation results.

The prototype displays the 160 held-out scenarios used for evaluation and demonstrates how the framework supports transparent, evidence-based security assessments.

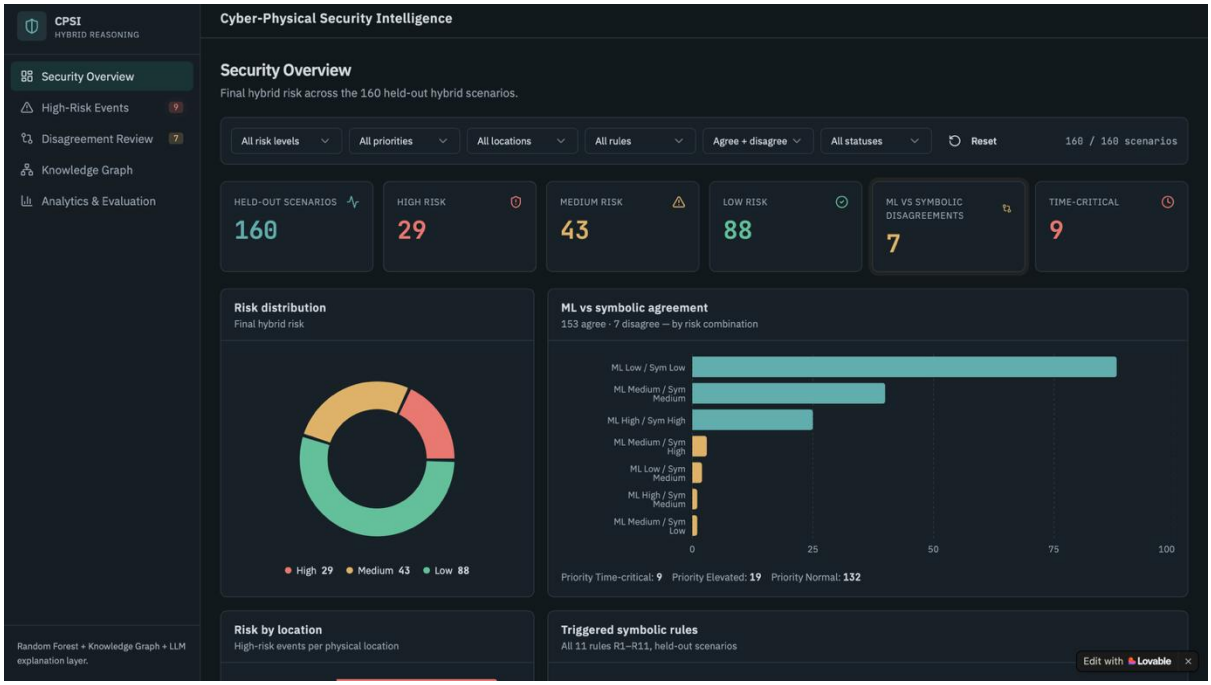


Figure 13. CPSI Dashboard – Security Overview.

10 Conclusion

In conclusion, this project developed a proof-of-concept hybrid reasoning framework that brings together cyber and physical security information to support contextual risk assessment. The Knowledge Graph represents relationships between users, devices, locations and events, while symbolic reasoning applies explicit security rules to identify suspicious activities and retain supporting evidence. Machine learning provides risk predictions, and the LLM translates the structured assessment outputs into natural-language explanations for analyst review.

The classification results demonstrate the benefit of combining cyber and physical features within the evaluated dataset. Hybrid features improved accuracy across all three classifiers. Random Forest, selected through training cross-validation, correctly classified 157 of 160 test scenarios, achieving 98.12% accuracy and identifying all 26 High-risk cases. Symbolic reasoning achieved 96.25% accuracy on the same test scenarios and provided traceable rule evidence, although some Low- and Medium-risk cases were incorrectly escalated to High risk.

The LLM evaluation demonstrated that structured reasoning outputs can support evidence-grounded explanations. All 30 manually reviewed explanations preserved the supplied risk assessment and supporting evidence, with no unsupported factual claims identified. However, clarity achieved a pass rate of 66.67%, indicating that explanation structure and readability require further refinement.

These findings are limited to scenarios with rubric-based reference labels, and explanation quality was assessed on a sample of the generated outputs. Future work should evaluate the framework using independently reviewed data, separate user groups or time periods, and multiple explanation reviewers. Within these limits, the project demonstrates a practical approach to combining classification, contextual relationships and explicit reasoning evidence while retaining human judgment as the final basis for security decisions.

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Appendix A – Project Proposal

Date of proposal: September 13, 2026

Project Title: Hybrid Machine Reasoning for Cyber-Physical Security

Group Members (name, Student ID):
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Sponsor/Client: *(Company Name, Address and Contact Name, Email, if any)*
N/A

Background/Aims/Objectives:

Background

Organizations commonly manage cybersecurity and physical security information in separate systems. Cybersecurity tools monitor events such as suspicious network traffic, authentication failures, and security alerts, while physical security systems record access to buildings and restricted locations. When these events are analyzed independently, potentially important connections may be missed.

For example, an employee may access a restricted server room outside normal working hours, and shortly afterwards, their assigned device may generate a suspicious network activity alert. Each event alone may not be sufficient to indicate a serious threat. However, when the events are linked to the same user and occur within a short time period, they may represent a significant cyber physical security risk.

This project proposes a hybrid machine reasoning framework that combines cyber and physical security events to identify such risks. The framework will use a Knowledge Graph to represent relationships between users, devices, locations, and events; rule-based reasoning to identify known suspicious patterns; and an interpretable classification model to classify scenarios as No Risk, Low, Medium, or High Risk. An LLM will generate a clear, evidence-based explanation of the result, but it will not make the final security decision.

Aim

The aim of this project is to develop a proof-of-concept hybrid machine reasoning system that correlates cyber and physical security events to detect cyber physical security risks that may not be visible when each domain is analyzed separately.

Objectives

- **Improve security risk detection:**
  Identify potential cyber physical security risks by analyzing cyber and physical access events together, rather than treating them as separate incidents.
- **Provide contextual security insights:**
  Reveal meaningful relationships between users, devices, locations, and timestamps to detect suspicious patterns that may otherwise be overlooked.
- ***Support* proactive risk management:**
  Classify scenarios as No Risk, Low, Medium, or High Risk to help security teams prioritize cases requiring attention or further investigation.
- **Provide transparent and explainable decisions:**
  Generate evidence-based explanations showing the relevant events, relationships, and rules that contributed to each risk classification.

- **Enhance analyst decision making:**
  Support security analysts with a combined rule-based and machine learning assessment, while keeping human review central for ambiguous or conflicting cases.

Project Descriptions:

Project Overview

This project proposes a hybrid machine-reasoning framework for detecting cyber-physical security risks. It addresses situations where cyber events and physical-access events may appear normal when analyzed separately but become suspicious when their relationship is considered. For example, access to a restricted server room outside working hours followed shortly by suspicious network activity from the same user's assigned device may indicate a high-risk security incident.

Proposed Solution and Approach

The proposed framework combines a Knowledge Graph, rule-based reasoning, and an interpretable classification model. The Knowledge Graph connects cyber and physical-security information, while security rules identify known suspicious patterns. Features extracted from the Knowledge Graph are then used by the Classification model to classify each scenario as No risk, Low, Medium, or High Risk. The system presents supporting evidence and triggered rules to make the result understandable to security analysts.

Data Sources and Dataset Structure

The project will build upon an existing cybersecurity dataset and extend it with synthetic organizational and physical-security information, it will use three related datasets: a Users' dataset, a Physical Events dataset, and a Cyber Events dataset. The datasets will be linked through common identifiers, particularly user_id, device_id, and event timestamps. This enables the framework to correlate a user's physical-access behaviour with cyber activity from their assigned device.

Users Dataset

Field	Description
User_id	Unique identifier assigned to each user. Used to link user information with physical and cyber events.
Role	Organizational role of the user, providing context about expected activities.
Department	Department to which the user belongs within the organization.
Device_id	Unique identifier of the device assigned to the user, enabling linkage with cyber events.
Normal_start	Expected start time of the user's normal working hours.
Normal_end	Expected end time of the user's normal working hours.

Physical Events Dataset

Field	Description
event_id	Unique identifier assigned to each physical-access event.
user_id	Identifier of the user associated with the physical-access event.
timestamp	Date and time at which the physical-access event occurred.
location	Physical location where the event occurred, such as an office, data centre, or server room.
Location_type	Classification of the location as normal or restricted.
access_result	Outcome of the physical-access attempt, recorded as granted or denied.

Cyber Events Dataset

Field	Description

Event_id	Unique identifier assigned to each cyber event.
User_id	Identifier of the user associated with the cyber event.
device_id	Identifier of the device associated with the cyber event.
Timestamp	Date and time at which the cyber event occurred.
Severity	Severity level of the cyber event: No risk, Low, Medium, or High.
Failed_logins	Number of unsuccessful authentication attempts associated with the event..
files_accessed	Number of files accessed during the cyber activity.
sensitive_files	Number of sensitive files accessed during the cyber activity.
usb_connected	Indicates whether a USB or removable-storage device was connected during the event (1 = Yes, 0 = No).

Key Workflow

1. Prepare and integrate data: Cybersecurity and physical-access datasets will be prepared and connected using user IDs, assigned devices, and timestamps.
2. Build the Knowledge Graph: The integrated data will be represented using the relationships User to Device to Cyber Event and User to Physical Access Event to Location. This enables the system to identify relationships between physical and cyber activities.
3. Apply rule-based reasoning: Security rules will identify suspicious patterns. For example, if a user accesses a restricted location outside working hours and their associated device shows suspicious activity shortly afterwards, the scenario will be flagged as high risk. This stage produces triggered rules and identified suspicious relationships.
4. Extract risk features: Relevant Knowledge Graph information will be converted into features, including after-hours access, restricted-location access, number of denied attempts, suspicious cyber activity, cyber-event severity, and the time difference between physical and cyber events.
5. Classify the security risk: The extracted features will be used as input to a Classification model to predict No risk, Low, Medium, or High Risk. For example, after-hours access combined with restricted-location access and suspicious cyber activity is expected to result in a High-Risk classification. The Classification model is used because its decision logic is interpretable.

Main Features and Functionalities

- Correlate physical-access and cybersecurity events in one unified security context.
- Identify suspicious relationships between users, devices, locations, and timestamps.
- Detect rule-based security patterns, such as after-hours access to restricted locations.
- Classify scenarios by risk level.
- Present supporting evidence and triggered rules for each classification.
- Flag uncertain or conflicting cases for security-analyst review.

Expected Deliverables

- Integrated cyber and synthetic physical-security dataset.
- Knowledge Graph representing users, devices, cyber events, access events, and locations.
- Rule-based reasoning component for suspicious-pattern detection.
- Risk classification model.
- Results display showing risk level, supporting evidence, and triggered rules.
- Evaluation report covering classification performance and reasoning correctness.

Scope and Limitations

This project is a proof of concept intended to support, rather than replace, security analysts. It will use available cybersecurity data and synthetic physical-access data because real organizational access-control data is sensitive and difficult to obtain. The framework will focus on detecting and explaining predefined cyber-physical risk scenarios rather than operating as a full real-time production security system.

Appendix B – Mapping System Functionalities to Intelligent Reasoning Systems Course

Table B1 summarises how the project applies knowledge, techniques and skills from Machine Reasoning (MR), Reasoning Systems (RS) and Cognitive Systems (CGS).

Table B1. Mapping of project functionalities to MR, RS and CGS

Course	System functionality	Knowledge and techniques applied	Skills demonstrated in the project
Machine Reasoning	Knowledge graph	Knowledge representation using entities, relationships and graph queries.	Build the Neo4j graph and retrieve linked users, devices, locations and events.
	Symbolic risk assessment	IF–THEN rules and logical inference.	Translate security conditions into rules and record the evidence supporting each risk assessment.
	Machine learning classification	Supervised classification using Decision Tree, Random Forest and Gradient Boosting.	Train and compare cyber-only, physical-only and hybrid models to predict Low, Medium and High risk.
	Model evaluation	Train/test splitting, cross-validation and classification metrics.	Tune model settings and compare accuracy, precision, recall, F1 and classification errors.
Reasoning Systems	Integration of system components	Hybrid reasoning combining graph context, symbolic rules and machine learning.	Connect component outputs for the same scenario and combine ML and symbolic risk assessments using a defined policy.
	Traceable decision support	Evidence-based reasoning and human oversight.	Retain risk predictions, triggered rules and supporting facts for analyst review.
Cognitive Systems	Natural-language explanations	Language generation and prompting with structured context.	Use the LLM to explain existing risk assessments using supplied rules and evidence.
	Explanation evaluation	Assessment of factual consistency and communication quality.	Review explanations for risk consistency, evidence faithfulness, unsupported claims and clarity.

Appendix C – CPSI Dashboard User Guide

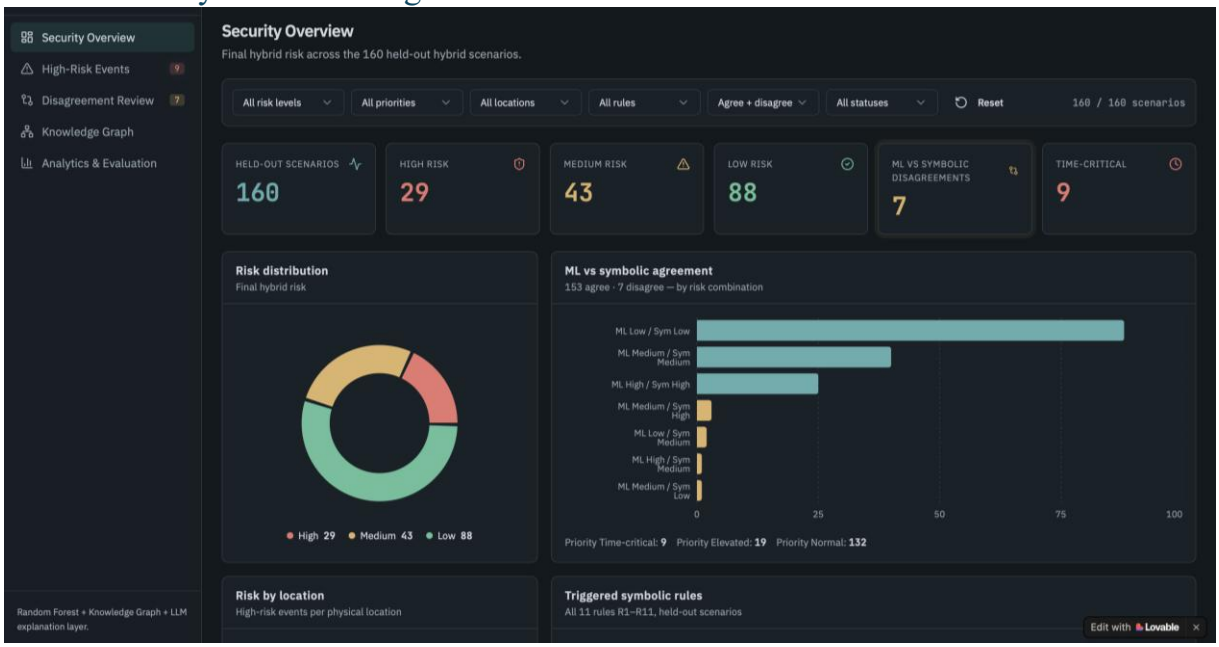
Getting started & security monitoring

Purpose. The dashboard supports review of cyber-physical security scenarios by combining Random Forest predictions, symbolic rules, knowledge graph (KG) relationships, and AI-generated explanations.

Access. Open https://pixel-perfect-match-8226.lovable.app and use the left sidebar to navigate between the five dashboard sections.

Page	Use it for
Security Overview	See risk, priority, location, agreement and rule summaries; open a priority scenario.
High-Risk Events	Browse all scenarios, not just High risk; select a row to investigate.
Disagreement Review	Review unresolved disagreements and a separate queue of open time-critical agreement cases.
Knowledge Graph	Explore one scenario's users, devices, events, locations and relationships.
Analytics & Evaluation	Compare the three methods, explanation-review results and disagreement patterns.

C.1 Security Overview Page



KPI cards. The current held-out view contains 160 scenarios: 29 High, 43 Medium and 88 Low risk; 7 ML–symbolic disagreements; and 9 time-critical alerts. Counts reflect the displayed dataset, not live security incidents.

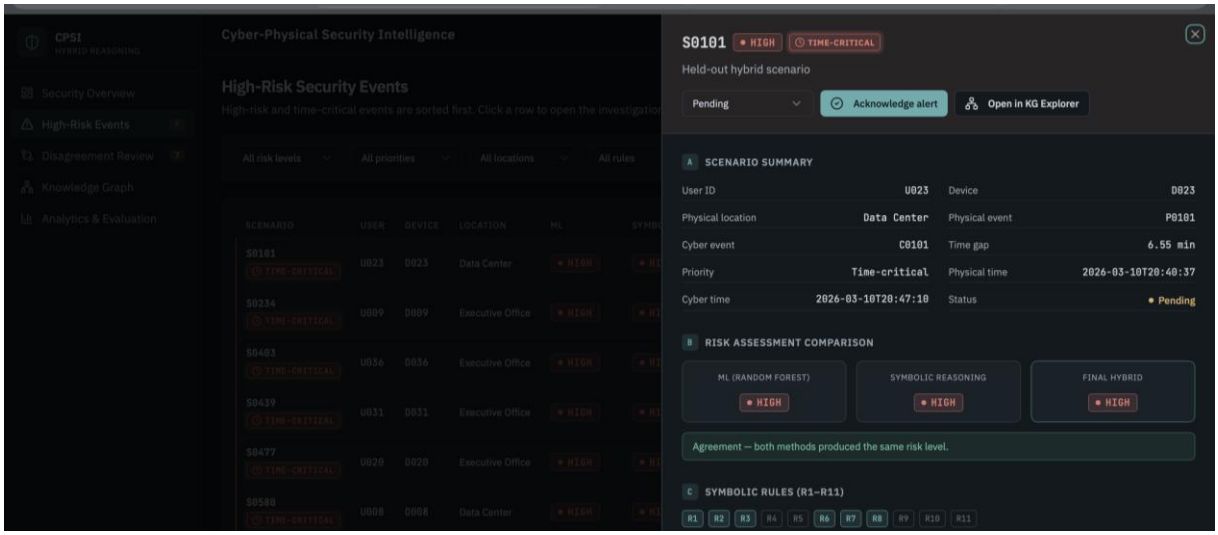
Filter the view. Use the six dropdowns to narrow by risk level, priority, location, rule, agreement and review status. The scenario count at the right shows the filtered population. Select Reset to clear filters.

Explore patterns. Review the risk distribution chart, ML versus symbolic agreement bars, risk by location and triggered-rule frequency. Scroll to Priority scenarios and select a row to investigate.

High-risk events and analyst workflow

C.2 Open an investigation

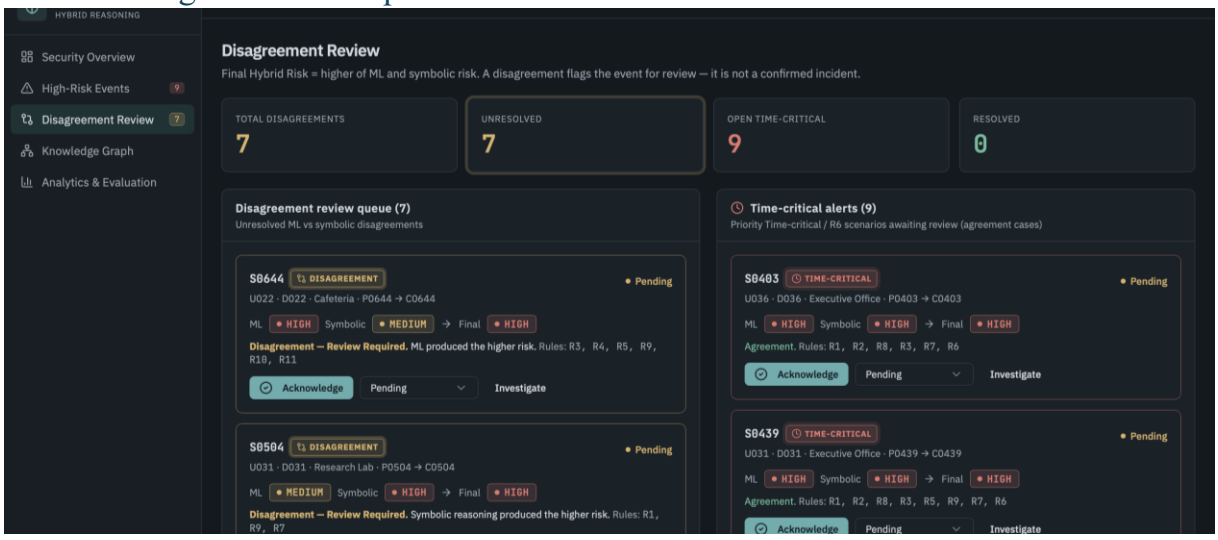
Find a case. Select High-Risk Events in the sidebar. Time-critical and High-risk scenarios appear first. The table can also show Medium/Low cases; select Risk → High to restrict it. Select a row to investigate.



Inspect the evidence. The panel presents (A) scenario details and event times; (B) Random Forest, symbolic and final hybrid risks; (C) triggered rules and the facts satisfying them; (D) KG validation; (E) the AI risk explanation; and (F) a relationship diagram farther down.

Interpret risk. The final hybrid risk is the higher of the ML and symbolic predictions. Time-critical indicates a cross-domain rule with a time gap of ten minutes or less; it is a prioritization flag, not confirmation of an incident.

C.3 Manage the review queue



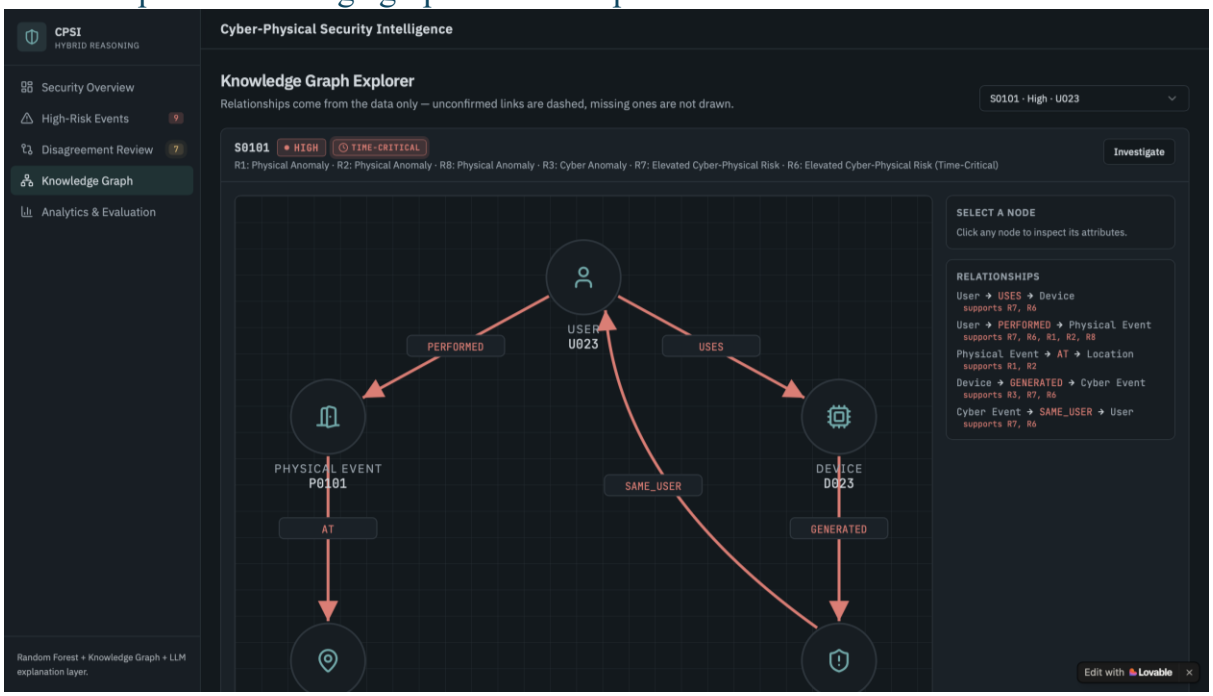
Review disagreement cases. Select Disagreement Review to inspect differing ML and symbolic classifications and open time-critical alerts. Select Investigate to review evidence. Acknowledge records

awareness, not resolution; set Under Review while assessing or Resolved when complete. Resolved disagreements leave the unresolved queue.

Navigate to the graph. From an open investigation, select Open in KG Explorer to examine linked user, device, physical-event, location and cyber-event relationships.

Trace evidence and interpret results

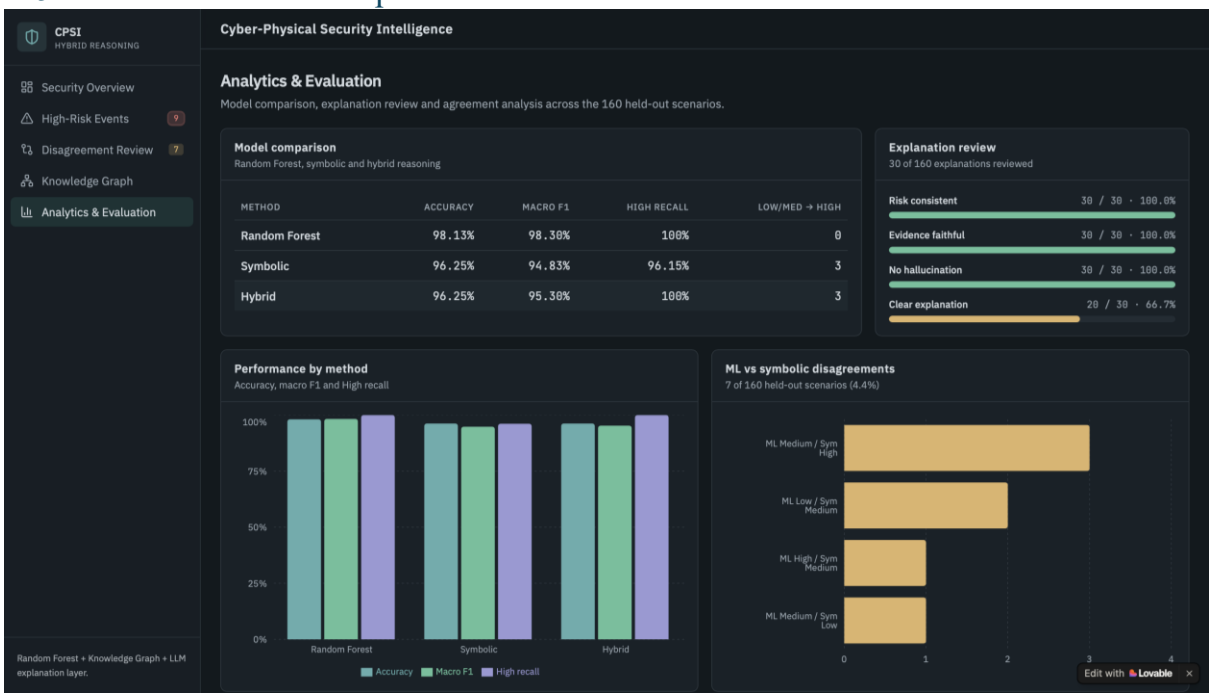
C.4 Explore knowledge graph relationships



Select a scenario. Open Knowledge Graph and choose a scenario from the top-right dropdown. The graph displays relationships among the user, assigned device, physical event, location and cyber event.

Inspect nodes and links. Select a node to view its attributes in the side panel. Review the Relationships panel to see which links support triggered rules. As noted in the interface, unconfirmed links are dashed and missing links are omitted.

C.5 Review framework performance



Open Analytics & Evaluation. Compare Random Forest, symbolic and hybrid methods using accuracy, Macro F1, High-risk recall and Low/Medium-to-High escalations. The page also shows ML–symbolic disagreement patterns and the manual LLM explanation review.

Read results correctly. On the 160 held-out scenarios, accuracy is 98.13% for Random Forest and 96.25% for symbolic and hybrid reasoning. Hybrid High-risk recall is 100%. Manual explanation review covers 30 of 160 cases: 30 passed consistency, faithfulness and no-hallucination checks; 20 passed clarity.

Where the displayed results come from

Source	Used for
hybrid_results.csv	Primary 160-scenario held-out hybrid dataset and final risk.
explanations.csv	LLM explanation joined to a scenario using scenario_id.
evaluation_metrics.csv	Random Forest, Symbolic and Hybrid evaluation metrics.
scenario_id_mapping.csv	Links each scenario to its source user, device and events.
completed LLM explanation review workbook	Manual review criteria and pass rates for 30 LLM explanations.
Review status storage	Acknowledgements and review statuses are saved in the current browser only, not shared across users or devices.